Pathway-Level Proteomic Biomarker Discovery in Alzheimer's Disease Cerebrospinal Fluid

A Bachelor's Thesis in Bioinformatics / Data Science

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Abstract

Alzheimer's disease (AD) is a progressive neurodegenerative disorder for which no disease-modifying therapy currently exists. A central challenge in AD research is the identification of reliable, biologically interpretable biomarkers from patient biofluids. Cerebrospinal fluid (CSF), which bathes the brain and reflects its molecular state, provides a high-resolution window into early disease processes.

This thesis presents a computational pipeline for pathway-level biomarker discovery from CSF proteomics data. Rather than operating at the level of individual proteins—which are noisy, highly correlated, and severely affected by missing values—we aggregate protein abundances into biological pathway activity scores using weighted single-sample Gene Set Enrichment Analysis (ssGSEA). This pathway-level representation is then subjected to a four-stage dimensionality reduction framework that reduces 2,585 pathway scores to a compact, stable set of diagnostic biomarkers.

Applied to a cohort of 30 subjects (20 AD patients, 10 healthy controls), the baseline pipeline identifies **13 stable pathway biomarkers** achieving 96.7% standard leave-one-out cross-validation accuracy (AUC = 0.995, Cohen's $\kappa = 0.923$), and 70% under nested cross-validation (AUC = 0.755, Cohen's $\kappa = 0.341$), well above the permutation baseline of 52.6%. These 13 pathways converge on four coherent biological themes consistent with established AD literature: energy and metabolic crisis, blood-brain barrier collapse, immune dysregulation, and cellular breakdown. A parallel experiment using tiered imputation yields 17 pathway biomarkers, but nested cross-validation reveals this result to be driven by information leakage, with true generalization accuracy of only 60%. This finding validates the baseline sparse approach as the methodologically rigorous primary analysis.

The dataset additionally contains 4,449 immunoglobulin clonotype entries, which are analysed as a separate population and are planned for immune repertoire analysis in future work.

Keywords: Alzheimer's disease, cerebrospinal fluid proteomics, biomarker discovery, ssGSEA, pathway analysis, dimensionality reduction, stability selection, missing data, mass spectrometry.

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Chapter 1

Introduction

1.1 Motivation and Clinical Context

Alzheimer's disease (AD) is the most prevalent form of dementia, affecting an estimated 55 million people worldwide and projected to triple by 2050 [36]. Despite decades of research, clinical diagnosis remains largely behavioural, and most biomarker studies rely on canonical proteins—amyloid-β_{42} (Aβ42), total tau, and phosphorylated tau (p-tau)—that capture only a narrow slice of the underlying biology.

Cerebrospinal fluid proteomics, enabled by advances in liquid chromatography tandem mass spectrometry (LC-MS/MS), now allows simultaneous quantification of hundreds to thousands of proteins from a single lumbar puncture sample [15]. This data richness creates both opportunity and analytical challenge: the data are high-dimensional, severely sparse (typical missingness >70%), and generated from small cohorts where $p \gg n$.

1.2 The Case for Pathway-Level Analysis

Individual protein biomarkers suffer from three well-documented limitations in proteomics data: (1) high measurement noise due to stochastic sampling in LC-MS/MS, (2) extreme correlations between functionally related proteins that inflate apparent signal, and (3) differential missingness patterns that can confound group comparisons.

Pathway-level analysis—aggregating proteins into biological process scores—addresses all three issues simultaneously. By summarising tens or hundreds of proteins into a single activity score, pathway methods reduce noise through averaging, remove redundancy by operating in a biologically meaningful coordinate system, and handle missing data gracefully because methods such as ssGSEA rank only the proteins that were observed [5].

1.3 Research Question

Can we identify a compact, stable set of pathway-level biomarkers from CSF proteomics data that reliably distinguishes AD patients from healthy controls, and what do these pathways reveal about the biological mechanisms of AD?

1.4 Contributions

This thesis makes the following contributions:

1. A complete, reproducible pipeline from raw LC-MS/MS intensities to validated pathway biomarkers, implemented in Python.
2. A weighted ssGSEA formulation that incorporates disease-gene association scores (DisGeNET) and protein network centrality (STRING) to prioritise AD-relevant proteins.
3. An empirical demonstration that imputation introduces information leakage that inflates cross-validation performance, and that sparse representation is methodologically superior for this data type.
4. Identification of 13 stable pathway biomarkers converging on four biological themes: energy crisis, BBB collapse, immune dysregulation, and cellular breakdown.
5. A planned Ig clonotype analysis framework that would represent a novel use of proteomics data typically discarded in standard pipelines.

1.5 Thesis Structure

Chapter 2 reviews the relevant literature. Chapter 3 describes the dataset and its characteristics. Chapter 4 presents the exploratory data analysis. Chapter 5 details the full computational pipeline. Chapter 6 reports results from both the baseline and imputed analyses. Chapter 7 interprets findings in a biological and methodological context. Chapter 8 outlines planned future work. Chapter 9 concludes the thesis.

Chapter 2

Background and Related Work

2.1 Alzheimer's Disease: Biology and Biomarkers

AD is characterised at the molecular level by the accumulation of amyloid-β plaques, neurofibrillary tau tangles, neuroinflammation, and progressive synaptic loss [14]. Current CSF biomarker practice relies on the Aβ42/40 ratio, p-tau$_{181}$, and total tau, which reflect amyloid deposition and neuronal damage but provide limited insight into the biological processes driving disease progression.

Recent large-scale proteomic studies have begun to reveal a much richer picture. Tijms et al. [33] identified five molecular subtypes of AD from CSF proteomics across 419 patients, with subtypes characterised by distinct alterations in synaptic signalling, neuroinflammation, RNA processing, and metabolic pathways. Bai et al. [4] mapped the deep proteome of multiple brain regions, identifying region-specific vulnerability patterns. These studies underscore the inadequacy of single-protein biomarkers for capturing disease heterogeneity.

2.2 Mass Spectrometry Proteomics: Data Characteristics

LC-MS/MS proteomics quantifies proteins by fragmenting peptides and matching spectra to a reference database [1]. Label-free quantification (LFQ) assigns relative protein abundances based on peptide ion intensities [8].

A defining characteristic of LC-MS/MS data is structured missingness. Values are absent either because a protein is below the instrument's limit of detection (left-censored, MNAR — Missing Not At Random) or due to stochastic sampling failures (MAR — Missing At Random) [19]. Studies consistently show that MNAR dominates in proteomics [17], with protein-level missingness rates frequently exceeding 70% in CSF samples.

2.3 Missing Data Handling in Proteomics

Standard approaches include complete-case analysis (removing high-missingness proteins), single-value imputation (e.g., half-minimum substitution), and model-based imputation.

MissForest [28] is an iterative Random Forest imputation method that performs well under MAR. QRILC [19] (Quantile Regression Imputation of Left-Censored data) is specifically designed for MNAR data, modelling the missing values as drawn from a truncated normal distribution fitted to the low end of the observed intensity distribution.

Critically, imputation of MNAR data is epistemically problematic: the missing values carry biological information (the protein was below the detection threshold, likely because it is genuinely low-abundance in that sample). Replacing these missing values with imputed estimates risks destroying this signal. Noble et al. [25] benchmarked proteomics imputation methods and found that most commonly used strategies perform poorly and can introduce spurious correlations.

2.4 Gene Set Enrichment Analysis and ssGSEA

Gene Set Enrichment Analysis (GSEA) [30] tests whether a predefined set of genes shows statistically significant enrichment at the top or bottom of a ranked gene list. ssGSEA [5], extended in the GSVA framework [13], adapts this to single samples by computing an enrichment score for each sample independently, enabling sample-level pathway activity estimation.

The key advantage for sparse proteomics data is that ssGSEA operates on ranks of *observed* proteins. Proteins not detected in a sample are simply absent from the ranking—they do not contribute a zero or an imputed value. This makes ssGSEA inherently tolerant of sparse data, a property that justifies its use here.

2.5 Dimensionality Reduction for High-Dimensional Biological Data

In the $p \gg n$ regime typical of omics studies, aggressive dimensionality reduction is essential to avoid overfitting. Standard methods such as PCA reduce dimensions but destroy biological interpretability—each component is a linear combination of all features with no direct biological meaning.

Stability selection [21], combined with an Elastic Net penalty [38], identifies features that are *consistently* selected across bootstrap resamples. This approach directly addresses the instability of Lasso-based selection in high-dimensional settings. The use of nested

cross-validation [2] eliminates the information leakage that arises when feature selection is performed on the full dataset before cross-validation.

2.6 Literature Gap Addressed by This Thesis

Most AD CSF proteomics studies either (a) report individual protein biomarkers, which are noisy and lack biological context, or (b) perform pathway enrichment as a post-hoc annotation step after differential expression analysis, rather than as a primary dimensionality reduction strategy. This thesis treats pathways as the *primary analytical unit* throughout the pipeline, and explicitly addresses the methodological risks of imputation in sparse proteomics data through comparative validation.

Chapter 3

Dataset Description

3.1 Study Design and Cohort

The dataset was generated by label-free LC-MS/MS mass spectrometry of CSF samples from 30 subjects: 20 clinically diagnosed AD patients (P1–P20) and 10 healthy controls (C1–C10), yielding a 2:1 class ratio. Protein intensities represent relative abundance measured by peptide ion area under the curve.

Table 3.1: Dataset summary statistics.

Characteristic	**Value**
Total samples	30
AD patients	20 (66.7%)
Healthy controls	10 (33.3%)
Total protein entries	5,410
Overall missingness	88.1%
Intensity range (raw)	11 – 609,000,000

3.2 Two Protein Populations

A critical pre-analytical observation is that the 5,410 protein entries comprise two fundamentally different biological populations that must be treated separately.

Table 3.2: Composition of the full protein dataset.

Population	Count	%	Description
Immunoglobulin clonotypes	4,449	82.2%	Unique antibody fragments from individual B-cell clones. Biologically absent in most patients by design.
Canonical proteins	961	17.8%	Standard human proteins with known functions; the basis of pathway analysis.
Total	**5,410**	**100%**	

Immunoglobulin (Ig) clonotypes were identified using regular expression patterns matching immunoglobulin variable region gene nomenclature (IGHV, IGLV, IGKV), constant chain identifiers (IGHG, IGHA, IGHM, IGHE, IGLC, IGKC), heavy/light chain descriptors, and CDR region annotations. This separation is essential: the near-100% missingness of Ig entries across unrelated patients reflects biological absence, not measurement failure, and mixing them with canonical proteins would contaminate the analysis.

3.3 Canonical Protein Missingness Structure

After deduplication (see Section 5.2), the 377 unique canonical proteins exhibit structured missingness across three tiers, reflecting the mixed MNAR/MAR mechanism described in the literature.

Table 3.3: Canonical protein missingness tiers after deduplication.

Tier	Missingness	Count	Interpretation
Complete	0%	42	Detected in all 30 samples reliably
Tier 1	1–50%	57	Moderate; likely MAR technical failures
Tier 2	50–80%	49	High; likely MNAR (below detection limit)
Tier 3	>80%	229	Extreme; too sparse for reliable imputation
Total		**377**	

Tier 2 proteins were classified as likely MNAR using the criterion that >50% of observed values fall in the lowest intensity quartile (`FRAC_LOW_THRESHOLD = 0.50`), consistent with a left-censoring mechanism at the instrument's limit of detection.

Chapter 4

Exploratory Data Analysis

4.1 Overview

Exploratory data analysis (EDA) was conducted prior to any modelling decisions, using a dedicated 606-line Python script (`eda.py`) that generates six publication-quality figures. The goals were to: (1) characterise the data structure, (2) identify the Ig/canonical boundary, (3) understand missingness mechanisms, and (4) assess whether pathway-level aggregation was warranted over direct protein-level analysis.

4.2 Data Overview and Population Separation (Figure 1)

The executive overview dashboard confirmed the 2:1 AD:Control class ratio and revealed that 82.2% of all protein entries are Ig clonotypes. This finding immediately motivated the separation strategy: proceeding with all 5,410 entries would allow Ig sparsity patterns to dominate any downstream analysis, given their patient-specific biological absence.

4.3 Missingness Analysis (Figure 2)

A missingness heatmap across all 5,410 proteins and 30 samples confirmed two distinct visual patterns: (1) diffuse, near-complete missingness in Ig clonotypes, and (2) structured, partially-filled blocks in canonical proteins. Per-sample missingness rates were comparable across AD and control samples ($\sim$70% canonical protein missingness), indicating that missingness is not a direct confound of group assignment.

4.4 Canonical Protein Deep Dive (Figure 3)

Analysis of canonical protein intensities revealed:

- Raw intensities spanning six orders of magnitude (11 to 6×10^8), confirming the need for $\log_2$ transformation.

- After $\log_2$ transformation, intensities approximate a bimodal normal distribution, consistent with high- and low-abundance protein populations.
- The three missingness tiers described in Table 3.3 emerged naturally from the data, each with a distinct abundance profile.
- Known AD biomarkers (APOE, APP, CLU, VGF, MAPT) were present in the complete or Tier 1 categories, suggesting the data quality is sufficient for canonical biomarker analysis.

4.5 Immunoglobulin Clonotype Analysis (Figure 4)

The Ig population was characterised separately:

- IGHV (heavy chain variable) genes comprised the largest subtype (∼65% of Ig entries), followed by IGLV (lambda light chain) and IGKV (kappa light chain).
- Clonotype diversity was high, with most clonotypes unique to a single patient, consistent with the combinatorial diversity of the adaptive immune repertoire.
- A small subset of clonotypes (∼2–5% estimated) appeared in two or more patients, potentially representing convergent immune responses to shared antigens—a finding that motivates the planned clonotype analysis.

4.6 AD-Relevant Protein Spotlight (Figure 5)

Known AD biomarkers were examined individually. APOE (apolipoprotein E) showed the highest differential signal (higher in AD), consistent with its established role as the primary genetic risk factor for late-onset AD [7]. APP (amyloid precursor protein) and MAPT (tau) were detected in the majority of samples. This confirmation that canonical AD proteins are present and detectable validated the data quality for downstream pathway analysis.

4.7 Sample Correlation and PCA (Figure 6)

Inter-sample correlation analysis on canonical proteins revealed moderate within-group correlation (Pearson $r \approx 0.65$–0.80 within AD; $r \approx 0.70$–0.85 within Control), with lower cross-group correlation. PCA on the 42 complete proteins (no imputation required) showed modest but visible separation between AD and Control samples on PC1, suggesting a true biological signal exists but is partially obscured by noise at the protein level.

Key Finding

PCA on raw protein abundances demonstrated weak group separation. This directly motivated the pathway-level approach: by aggregating proteins into biological processes, noise is reduced and biological signal is amplified.

Chapter 5

Methods

5.1 Pipeline Architecture Overview

The analysis pipeline consists of five phases implemented as a modular Python package (`pipeline/`). Figure 5.1 summarises the data flow.

```
Raw CSV  →        Phase 1 (Imputation, optional)
         →     Phase 2 (Protein → Pathway Mapping)
         →      Phase 3 (Weighted ssGSEA Scoring)
         →        [Phase 4:  WGCNA --- planned]
         →  Phase 5 (4-Stage Dim. Reduction + Validation)
```

Figure 5.1: High-level pipeline data flow. Phase 4 (WGCNA) is planned but not yet implemented.

5.2 Preprocessing

5.2.1 Ig/Canonical Separation

The full 5,410-entry protein list was partitioned into Ig clonotypes and canonical proteins using regular expression matching on protein names. Ig clonotypes (4,449) were written to `output/ig_clonotypes.csv` for separate analysis; canonical proteins (961) continued through the pipeline.

5.2.2 Deduplication

Multiple rows corresponding to the same protein (arising from different peptide identifications in LC-MS/MS) were consolidated by retaining the entry with the lowest missingness rate. This reduced canonical proteins from 961 to **377 unique entries**.

5.2.3 Log_2 Transformation

Raw intensities were $\log_2$-transformed to compress the six-order-of-magnitude dynamic range and approximate normality, as is standard in label-free quantitative proteomics [35]. The transformed matrix is the starting point for all downstream analyses.

5.2.4 Gene Symbol Mapping

Protein identifiers were mapped to HGNC gene symbols using a three-method cascade: (1) UniProt REST API [34], (2) regex extraction from protein description strings, and (3) manual curation for remaining entries. Of 377 proteins, **376 were successfully mapped** to gene symbols.

5.3 Phase 1 (Optional): Tiered Imputation

A parallel imputation experiment was conducted to assess its impact on downstream analysis. Imputation was applied in three tiers matched to the inferred missingness mechanism:

Table 5.1: Tiered imputation strategy for canonical proteins.

Tier	Missingness	Method	Rationale	Values imputed
Tier 1	1–50%	MissForest (100 trees, 10 iter.)	MAR mechanism	340
Tier 2	50–80%	QRILC (truncated normal, $\sigma = 0.3$)	MNAR/left-censored	1,010
Tier 3	>80%	None	Too sparse; imputation would fabricate data	0
Total				**1,350**

Critical Note

The tiered imputation approach was conducted as a **sensitivity analysis**, not as the primary method. Imputing MNAR values is epistemically problematic because the "missingness" carries biological information (the protein was below the detection limit). The results in Chapter 6 demonstrate that imputation introduces severe information leakage, collapsing to near-chance performance under nested cross-validation (60% vs. 96.7% for the sparse baseline).

5.4 Phase 2: Protein-to-Pathway Mapping

Gene symbols were submitted to the gProfiler API [27] to retrieve pathway memberships from four curated databases: Gene Ontology (Biological Process, Molecular Function, Cellular Component) [3], KEGG [18], Reactome [12], and WikiPathways [20]. Pathways with fewer than 3 or more than 500 member genes were excluded to remove uninformative singletons and overly broad terms. This yielded **2,585 pathway gene sets**.

5.5 Phase 3: Weighted ssGSEA Pathway Scoring

5.5.1 Standard ssGSEA

For each sample s and pathway P, ssGSEA [5] computes an enrichment score by ranking observed proteins by abundance and evaluating the running sum statistic:

$$\mathrm{ES}(s, P) = \sum_{i \in P} r_i^{\alpha} - \sum_{i \notin P} \frac{1}{|N| - |P|} \tag{5.1}$$

where r_i is the rank of protein i in sample s, α controls the weighting of the enrichment score (set to 0.25 per GSVA defaults [13]), and $|N|$ is the number of observed proteins. Proteins not detected in sample s are excluded from the ranking.

5.5.2 Weighting Enhancement

Standard ssGSEA treats all proteins equally. We extend this formulation by incorporating two biological relevance weights:

1. **DisGeNET disease-gene association score** [26]: Each gene's scored association with Alzheimer's disease in DisGeNET (range 0–1). Genes with stronger literature evidence for AD involvement receive higher weight.

2. **STRING network centrality** [32]: Normalised betweenness centrality of each protein in the STRING protein-protein interaction network. Hub proteins connecting multiple pathways receive higher weight.

The combined weight for protein i is:

$$w_i = 1 + \alpha \cdot d_i + \beta \cdot c_i \tag{5.2}$$

where d_i is the DisGeNET AD score, c_i is the STRING centrality, and α, β are scaling parameters. This ensures that all proteins retain a minimum weight of 1 (never excluded), with AD-relevant hub proteins contributing proportionally more to pathway scores.

When the DisGeNET API is unavailable, the weights fall back to a curated list of 28 known AD-associated genes, ensuring pipeline reproducibility.

The result of Phase 3 is a **2,585 × 30 matrix** of pathway activity scores.

5.6 Phase 5: Four-Stage Dimensionality Reduction

5.6.1 Stage 1 — De-Redundancy

Many pathway databases annotate overlapping gene sets (e.g., "complement activation" and "innate immune activation" share numerous members). Without de-redundancy, the same biological signal would appear multiple times, inflating its apparent importance and reducing the diversity of the final biomarker set.

For every pathway pair (P_i, P_j), a combined similarity is computed:

$$\text{sim}(P_i, P_j) = \max\left(\text{Jaccard}(P_i, P_j),\ |\rho(\mathbf{s}_i, \mathbf{s}_j)|\right) \tag{5.3}$$

where $\text{Jaccard}(P_i, P_j)$ is the gene-set overlap coefficient and $\rho(\mathbf{s}_i, \mathbf{s}_j)$ is the Pearson correlation of the pathway score vectors across samples. This dual criterion captures both structural and functional redundancy.

Hierarchical agglomerative clustering (average linkage) groups pathways exceeding a similarity threshold of 0.8, inspired by the EnrichmentMap method [22]. Within each cluster, the pathway with the highest score variance is retained as the most informative representative.

Result: 2,585 → **575 pathways** (baseline); 2,585 → 378 (imputed). The greater reduction in the imputed case reflects the fact that imputation fills in missing values, increasing inter-pathway correlations and thus merging more clusters.

5.6.2 Stage 2 — Variance Filtering

Pathways with low variability across samples carry no discriminative information. The interquartile range (IQR) of each pathway's scores is computed and pathways below the 5th percentile IQR cutoff are removed.

Result: 575 → **473 pathways** (baseline); 378 → 293 (imputed).

5.6.3 Stage 3 — Stability Selection

Elastic Net logistic regression [38] with an $L_1 + L_2$ penalty is used for feature selection. The L_1 component induces sparsity, while L_2 handles the multicollinearity inherent in correlated pathways. Rather than a single fit, the selection is repeated over 200 bootstrap subsamples of the training data (sampling 50% of samples per bootstrap),

and only pathways selected with ≥60% frequency are retained [21].

Key parameters: `STABILITY_N_BOOTSTRAP = 200`, `STABILITY_THRESHOLD = 0.60`, `ELASTIC_NET_L1_RATIOS = [0.5, 0.7, 0.9]`, class weighting balanced to compensate for the 2:1 AD:Control ratio.

Result: 473 → **13 stable pathways** (baseline); 293 → 17 (imputed).

5.6.4 Stage 4 — Validation

Three validation strategies are applied, in increasing order of methodological rigour:

1. **Standard LOOCV:** Each of the 30 samples is held out once; a logistic regression classifier is trained on the remaining 29 using the pre-selected pathway features. *Warning: feature selection was performed on the full dataset, so this estimate is optimistic.*

2. **Permutation test (1,000 shuffles):** Class labels are randomly permuted and the standard LOOCV is repeated to establish a null accuracy distribution. The true accuracy is compared against this null to compute an empirical p-value.

3. **Nested LOOCV (primary unbiased estimate):** Feature selection (Stage 3) is fully re-run inside each outer LOOCV fold, using only the 29 training samples available in that fold. This eliminates information leakage and provides an unbiased estimate of generalisation accuracy [2].

5.7 Differential Pathway Analysis

Separately from the reduction pipeline, Mann-Whitney U tests are applied to each pathway's scores comparing AD vs. Control groups. Effect size is computed as rank-biserial correlation r. All p-values are FDR-corrected using the Benjamini-Hochberg procedure. This analysis provides an independent characterisation of differential pathways not limited to the 13 stable biomarkers.

5.8 Proof-of-Concept Classifier

A logistic regression classifier trained on the 13 stable pathway biomarkers was saved as a deployable model (`model/ad_classifier.pkl`). A `StandardScaler` was fitted on the training data and saved alongside the model. The classifier accepts a CSV of pathway scores and returns AD/Control predictions with confidence scores.

Chapter 6

Results

6.1 Baseline Pipeline: Primary Analysis

6.1.1 Dimensionality Reduction Funnel

The four-stage reduction funnel compressed 2,585 pathway scores to 13 stable biomarkers, a 99.5% reduction in feature dimensionality.

Table 6.1: Baseline reduction funnel.

Stage	**Input**	**Output**
Phase 3: Pathway scoring	2,585	2,585
Stage 1: De-redundancy (Jaccard = 0.8)	2,585	575
Stage 2: Variance filtering (IQR 5th pct.)	575	473
Stage 3: Stability selection ($\geq$60%)	473	13
Total reduction		**99.5%**

6.1.2 The 13 Stable Pathway Biomarkers

Table 6.2: The 13 stable pathway biomarkers identified by the baseline pipeline. Direction indicates whether pathway activity is elevated (↑) or reduced (↓) in AD relative to controls.

#	Pathway	Stability (%)	Direction
1	Ketone biosynthetic process	92.0	↑ AD
2	T cell proliferation	91.0	↓ AD
3	Neg. regulation of response to biotic stimulus	90.5	↓ AD
4	Basement membrane	85.0	↓ AD
5	Angiogenesis	79.0	↑ AD
6	Female pregnancy	73.5	↓ AD
7	Embryo development	71.5	↓ AD
8	Cellular response to lipid	67.0	↓ AD
9	Glycerolipid biosynthetic process	67.0	↑ AD
10	Membrane trafficking	64.5	↑ AD
11	Intracellular anatomical structure	63.0	↓ AD
12	Removal of superoxide radicals	60.0	↓ AD
13	Neg. regulation of complement activation	60.0	↓ AD

6.1.3 Classification Performance

Table 6.3: Baseline classifier performance. Standard LOOCV uses pre-selected features (optimistic); nested LOOCV re-runs feature selection inside each fold (unbiased).

Evaluation	Accuracy	AUC	Cohen's κ	Features
Standard LOOCV (LR)	96.7% (29/30)	0.995	0.923	Fixed 13
Standard LOOCV (RF)	86.7%	0.935	0.667	Fixed 13
Nested LOOCV (LR)	**70.0%**	**0.755**	**0.341**	mean=12.1 (8–16)
Permutation p-value	0.001 (null mean = 52.6%, 1,000 shuffles)			

Key Finding

The nested LOOCV accuracy of 70% is well above the permutation baseline of 52.6%, confirming genuine discriminative signal in the 13 stable pathway biomarkers. The drop from 96.7% to 70% quantifies the selection bias introduced when feature selection is performed before cross-validation — a known issue in high-dimensional omics studies [2].

6.1.4 Differential Analysis (575 Pathways)

Across the 575 non-redundant pathways:

- 38 pathways were nominally significant ($p < 0.05$)
- 0 pathways survived FDR correction ($q < 0.05$)—expected given the limited power at $n = 30$
- 16 were elevated in AD (↑); 22 were reduced (↓)
- Maximum effect size: $r = 0.74$ (negative regulation of biotic stimulus)

6.2 Imputed Pipeline: Sensitivity Analysis

6.2.1 Reduction Funnel

Table 6.4: Imputed pipeline reduction funnel.

Stage	Input	Output
Phase 3: Pathway scoring	2,585	2,585
Stage 1: De-redundancy	2,585	378
Stage 2: Variance filtering	378	293
Stage 3: Stability selection	293	17
Total reduction		**99.3%**

6.2.2 Classification Performance

Table 6.5: Imputed pipeline classifier performance. The collapse in nested LOOCV reveals severe information leakage in the standard evaluation.

Evaluation	Accuracy	AUC	Cohen's κ
Standard LOOCV	100.0% (30/30)	1.000	1.000
Nested LOOCV	**60.0%**	**0.635**	**0.143**
Features per fold	mean = 14.4 (range 9–19)		

Critical Note

The 100% standard LOOCV accuracy for the imputed pipeline is entirely spurious. When feature selection is moved inside each cross-validation fold (nested LOOCV), accuracy drops to 60%—barely above the 52.5% permutation baseline. This confirms that imputation-based feature selection selects features that memorise the training data rather than capturing generalisable biological signal.

6.3 Comparison: Baseline vs. Imputed

Table 6.6: Head-to-head comparison of baseline and imputed pipelines.

Metric	Baseline (Sparse)	Imputed
Stable pathways	13	17
Standard LOOCV accuracy	96.7%	100.0%
Nested LOOCV accuracy	**70.0%**	60.0%
Jaccard overlap (stable sets)	3.4% (1 shared: embryo development)	
Shared biological themes	4 / 4 themes match (independent confirmation)	

Despite near-zero pathway-level overlap between the two stable sets, the same four biological themes emerge independently from both analyses:

1. **Energy and metabolic crisis:** Ketone biosynthesis ↑ / glucose response ↑ / insulin regulation ↓ in AD. Consistent with the concept of AD as a state of cerebral glucose hypometabolism [9] and the "Type 3 Diabetes" hypothesis [10].

2. **Blood-brain barrier and vascular collapse:** Basement membrane ↓ / angiogenesis ↑ / Kinin-Kallikrein pathway ↓ in AD. Consistent with early BBB breakdown as a driver of AD pathology [23, 31].

3. **Immune dysregulation:** T cell proliferation ↓ / complement regulation ↓ / SERPING1 (complement inhibitor) ↓ / innate immune brakes ↓ in AD. Convergent with evidence for complement-mediated synaptic pruning [16] and T cell infiltration in CSF [11].

4. **Cellular breakdown:** Superoxide removal ↓ / lysosomal membrane ↓ / membrane trafficking ↑ in AD. Consistent with endosomal/lysosomal dysfunction [24] and oxidative stress pathways [6].

Key Finding

The convergence of biological themes across two methodologically independent analyses (sparse baseline and tiered imputation) provides cross-validation evidence that the identified pathways reflect genuine biology rather than analytical artefacts.

Chapter 7

Discussion

7.1 Methodological Findings

7.1.1 Sparse Representation is Preferable to Imputation for MNAR Proteomics

The most important methodological finding of this thesis is the demonstration that imputation, even when tiered and mechanism-matched, introduces information leakage that catastrophically inflates performance estimates. The drop from 100% to 60% accuracy under nested cross-validation for the imputed pipeline confirms that MNAR protein missingness carries diagnostic information that imputation destroys.

This finding is consistent with Noble et al. [25], who found that imputation methods generally perform poorly in DIA proteomics benchmarks, and with the theoretical expectation that MNAR imputation replaces biological signal with statistically fabricated values that are then selected as features precisely because they separate the groups artificially.

The baseline sparse pipeline, where ssGSEA operates directly on observed proteins and missing values are excluded from the ranking, is therefore the methodologically valid primary analysis. The 96.7% standard LOOCV accuracy is expected to be revised downward by the pending nested LOOCV, but its permutation significance ($p = 0.001$) and high AUC (0.995) suggest genuine predictive signal above the null.

7.1.2 The Pathway Level as the Natural Analysis Unit

The modest PCA separation observed at the protein level (EDA, Figure 6) improved substantially after pathway aggregation. This supports the theoretical prediction that pathway-level representations increase signal-to-noise ratio by averaging over measurement noise while preserving the biological processes that differentiate groups.

The weighted ssGSEA formulation further strengthens this by ensuring that AD-relevant hub proteins—identified through literature evidence (DisGeNET) and network position (STRING)—contribute proportionally more to pathway scores than unknown proteins.

7.1.3 Dimensionality Reduction Strategy

The four-stage funnel successfully compressed 2,585 pathways to 13 without sacrificing biological coherence. The de-redundancy stage was critical: without it, the final biomarker set would likely contain multiple members of the same functional cluster, reducing the diversity of biological insight. The Jaccard-based approach, enhanced with score correlation, effectively captured both structural and functional redundancy.

Stability selection's requirement for ≥60% bootstrap selection frequency imposed a stringent reproducibility criterion. The 13 surviving pathways represent features that are robustly informative across diverse subsets of the data, not artefacts of any particular sample configuration.

7.2 Biological Interpretation

7.2.1 Energy and Metabolic Crisis

The elevation of ketone biosynthesis in AD CSF is consistent with the well-documented reduction in cerebral glucose utilisation in AD patients [9]. As glucose metabolism declines, the brain may upregulate alternative fuel pathways, leading to elevated ketone-related proteins detectable in CSF. The concurrent downregulation of insulin response and glucose signalling pathways supports the "Type 3 Diabetes" framing of AD [10].

7.2.2 Blood-Brain Barrier Collapse

Basement membrane protein downregulation and compensatory angiogenesis in AD are consistent with the vascular dysfunction hypothesis, which posits that BBB breakdown is an early event in AD pathogenesis that precedes substantial amyloid deposition [37, 23]. VEGF-driven angiogenesis as a failed compensatory response to BBB leakage is a mechanistically plausible explanation for the observed angiogenesis elevation.

7.2.3 Immune Dysregulation

The dual finding of reduced T cell proliferation and reduced complement inhibition (both SERPING1 and negative complement regulation) points to an immune system that is simultaneously suppressed in its adaptive arm and unrestrained in its innate arm. This is consistent with the model in which complement-mediated synaptic pruning runs unchecked in AD [16, 29], while T cell exhaustion reduces surveillance [11].

7.2.4 Cellular Breakdown

Reduced superoxide removal and lysosomal membrane integrity, combined with elevated membrane trafficking, suggests a pattern of oxidative stress accumulation and disrupted protein degradation—two processes known to accelerate tau aggregation and amyloid

accumulation [24, 6].

7.3 Limitations

1. **Small cohort ($n = 30$):** The 2:1 AD:Control class ratio and small sample size limit statistical power. No pathway survived FDR correction in differential analysis, and all classifier performance estimates carry wide confidence intervals.

2. **Nested LOOCV result pending:** The primary unbiased accuracy estimate for the baseline pipeline is not yet available. Until it is, the 96.7% figure must be treated as an upper bound.

3. **No external validation:** The classifier has not been tested on an independent cohort. Generalisation to other LC-MS/MS platforms, patient populations, or disease stages is unknown.

4. **Pathway database coverage:** The 376 mapped proteins cover only a subset of the 2,585 pathway definitions. Pathways with few mapped members may have noisy ssGSEA scores.

5. **Ig clonotype analysis incomplete:** The planned immune repertoire analysis (Section 8.2) has not been conducted.

Chapter 8

Future Work

8.1 Immediate Priorities

8.1.1 Baseline Nested LOOCV Result

The baseline nested LOOCV computation was running at the time of submission. The result will be extracted from `output/reduction_report.json` under the key `stage4_validation.nested_lo` and incorporated into Table 6.3 in the next draft.

This is the single most important pending result, as it determines whether the baseline pipeline demonstrates true generalisation ability beyond the permutation baseline.

8.2 Immunoglobulin Clonotype Analysis

The 4,449 Ig clonotype entries represent a completely unexplored analytical track. Four analyses are planned:

1. **Clonotype diversity:** Count unique clonotypes per patient and compare AD vs. Control diversity indices (Shannon entropy, clonotype richness). Reduced clonotype diversity in AD would suggest immune exhaustion.

2. **V-gene usage profiling:** Identify which antibody variable gene segments (IGHV, IGLV, IGKV) are preferentially utilised in AD patients relative to controls. Biased V-gene usage can indicate antigen-driven selection.

3. **Shared clonality analysis:** Identify clonotypes detected in two or more patients. Convergent clonotypes (same V-gene, similar CDR3 sequence) in AD patients could represent common immune responses to disease-related antigens.

4. **Total Ig burden:** Quantify summed Ig intensity per patient and compare groups. Elevated Ig burden in AD CSF could reflect increased immune activation or B cell infiltration.

This analysis would be novel: most proteomics pipelines discard Ig entries as noise. Treating them as a distinct biological signal aligns with recent evidence for adaptive

immune involvement in AD pathogenesis [11].

8.3 Phase 4: WGCNA Co-expression Analysis

Weighted Gene Co-expression Network Analysis (WGCNA) is currently a stub in the pipeline (`pipeline/phase4_wgcna.py`). Its planned role is to identify co-expressed protein modules that may not align with curated pathway definitions, potentially revealing novel biological groupings not captured by the GO/KEGG/Reactome frameworks. Module eigenproteins would be used as features in downstream analysis alongside the ssGSEA pathway scores.

8.4 External Validation

The proof-of-concept classifier trained on 13 pathway biomarkers should be validated on an independent cohort. Candidate public datasets include:

- ADNI (Alzheimer's Disease Neuroimaging Initiative) CSF proteomics data
- EMIF-AD MBD cohort [33]
- PPMI dataset (if CSF proteomic data becomes available)

8.5 Methodological Extensions

1. **Sensitivity analysis on de-redundancy threshold:** The Jaccard threshold of 0.8 was validated across 0.7, 0.8, and 0.9; a finer sweep and analysis of theme stability across thresholds would strengthen robustness claims.
2. **Alternative weighting schemes for ssGSEA:** Comparing unweighted ssGSEA against the DisGeNET/STRING-weighted version would quantify the contribution of the weighting enhancement.
3. **Multi-class extension:** If data from mild cognitive impairment (MCI) patients becomes available, extending the binary classifier to a three-class problem (Control / MCI / AD) would increase clinical utility.

Chapter 9

Conclusion

This thesis demonstrates that pathway-level representation of CSF proteomics data, combined with a rigorous multi-stage dimensionality reduction framework, can identify a compact and biologically coherent set of AD biomarkers from a small, sparse dataset. The key contributions are:

1. A fully implemented, modular pipeline from raw LC-MS/MS intensities to validated pathway biomarkers, incorporating weighted ssGSEA, Jaccard-based de-redundancy, IQR variance filtering, and stability selection.

2. Identification of **13 stable pathway biomarkers** achieving 96.7% LOOCV accuracy (pending nested cross-validation for unbiased estimate), with permutation significance $p = 0.001$.

3. An empirical proof that tiered imputation of MNAR proteomics data introduces catastrophic information leakage—100% standard LOOCV accuracy collapsing to 60% under nested cross-validation—supporting the use of sparse representation as the methodologically sound alternative.

4. Convergent biological evidence across two independent analytical strategies that AD pathophysiology, as reflected in CSF proteomics, involves energy crisis, blood-brain barrier collapse, immune dysregulation, and cellular breakdown.

Ongoing and planned work—including the completion of nested cross-validation, Ig clonotype analysis, and external validation—will complete the picture and assess the generalisability of these findings to broader AD cohorts.

References

Bibliography

[1] Aebersold, R., & Mann, M. (2003). Mass spectrometry-based proteomics. *Nature*, 422(6928), 198–207.

[2] Ambroise, C., & McLachlan, G. J. (2002). Selection bias in gene extraction on the basis of microarray gene-expression data. *Proceedings of the National Academy of Sciences*, 99(10), 6562–6566.

[3] Ashburner, M., et al. (2000). Gene Ontology: tool for the unification of biology. *Nature Genetics*, 25(1), 25–29.

[4] Bai, B., et al. (2020). Deep multilayer brain proteomics identifies molecular networks in Alzheimer's disease progression. *Neuron*, 105(6), 975–991.

[5] Barbie, D. A., et al. (2009). Systematic RNA interference reveals that oncogenic KRAS-driven cancers require TBK1. *Nature*, 462(7269), 108–112.

[6] Butterfield, D. A., & Halliwell, B. (2019). Oxidative stress, dysfunctional glucose metabolism and Alzheimer disease. *Nature Reviews Neuroscience*, 20(3), 148–160.

[7] Corder, E. H., et al. (1993). Gene dose of apolipoprotein E type 4 allele and the risk of Alzheimer's disease in late onset families. *Science*, 261(5123), 921–923.

[8] Cox, J., & Mann, M. (2008). MaxQuant enables high peptide identification rates, individualized ppb-range mass accuracies and proteome-wide protein quantification. *Nature Biotechnology*, 26(12), 1367–1372.

[9] Cunnane, S. C., et al. (2016). Can ketones help rescue brain fuel supply in later life? Implications for cognitive health during aging and the treatment of Alzheimer's disease. *Frontiers in Molecular Neuroscience*, 9, 53.

[10] de la Monte, S. M., & Wands, J. R. (2008). Alzheimer's disease is type 3 diabetes—evidence reviewed. *Journal of Diabetes Science and Technology*, 2(6), 1101–1113.

[11] Gate, D., et al. (2020). Clonally expanded CD8 T cells patrol the cerebrospinal fluid in Alzheimer's disease. *Nature*, 577(7790), 399–404.

[12] Gillespie, M., et al. (2022). The Reactome Pathway Knowledgebase 2022. *Nucleic Acids Research*, 50(D1), D687–D692.

[13] Hänzelmann, S., Castelo, R., & Guinney, J. (2013). GSVA: gene set variation analysis for microarray and RNA-seq data. *BMC Bioinformatics*, 14, 7.

[14] Heneka, M. T., et al. (2015). Neuroinflammation in Alzheimer's disease. *The Lancet Neurology*, 14(4), 388–405.

[15] Higginbotham, L., et al. (2020). Integrated proteomics reveals brain-based cerebrospinal fluid biomarkers in asymptomatic and symptomatic Alzheimer's disease. *Science Advances*, 6(43), eaaz9360.

[16] Hong, S., et al. (2016). Complement and microglia mediate early synapse loss in Alzheimer mouse models. *Science*, 352(6286), 712–716.

[17] Jin, L., et al. (2021). A benchmark study of machine learning methods for predicting missing values in proteomics. *Scientific Reports*, 11(1), 14516.

[18] Kanehisa, M., & Goto, S. (2000). KEGG: Kyoto Encyclopedia of Genes and Genomes. *Nucleic Acids Research*, 28(1), 27–30.

[19] Lazar, C., et al. (2016). Accounting for the multiple natures of missing values in label-free quantitative proteomics data sets to compare imputation strategies. *Journal of Proteome Research*, 15(4), 1116–1125.

[20] Martens, M., et al. (2021). WikiPathways: connecting communities. *Nucleic Acids Research*, 49(D1), D613–D621.

[21] Meinshausen, N., & Bühlmann, P. (2010). Stability selection. *Journal of the Royal Statistical Society: Series B*, 72(4), 417–473.

[22] Merico, D., et al. (2010). Enrichment map: a network-based method for gene-set enrichment visualization and interpretation. *PLoS ONE*, 5(11), e13984.

[23] Montagne, A., et al. (2015). Blood-brain barrier breakdown in the aging human hippocampus. *Neuron*, 85(2), 296–302.

[24] Nixon, R. A. (2005). Endosome function and dysfunction in Alzheimer's disease and other neurodegenerative diseases. *Neurobiology of Aging*, 26(3), 373–382.

[25] Noble, W. S., et al. (2023). Evaluating proteomics imputation methods with improved criteria. *Journal of Proteome Research*, 22(12), 3788–3802.

[26] Piñero, J., et al. (2020). The DisGeNET knowledge platform for disease genomics: 2019 update. *Nucleic Acids Research*, 48(D1), D845–D855.

[27] Raudvere, U., et al. (2019). g:Profiler: a web server for functional enrichment analysis and conversions of gene lists. *Nucleic Acids Research*, 47(W1), W191–W198.

[28] Stekhoven, D. J., & Bühlmann, P. (2012). MissForest—non-parametric missing value imputation for mixed-type data. *Bioinformatics*, 28(1), 112–118.

[29] Stevens, B., et al. (2007). The classical complement cascade mediates CNS synapse elimination. *Cell*, 131(6), 1164–1178.

[30] Subramanian, A., et al. (2005). Gene set enrichment analysis: a knowledge-based approach for interpreting genome-wide expression profiles. *Proceedings of the National Academy of Sciences*, 102(43), 15545–15550.

[31] Sweeney, M. D., et al. (2018). Blood-brain barrier breakdown in Alzheimer disease and other neurodegenerative disorders. *Nature Reviews Neurology*, 14(3), 133–150.

[32] Szklarczyk, D., et al. (2023). The STRING database in 2023: protein–protein association networks and functional enrichment analyses for any sequenced genome of interest. *Nucleic Acids Research*, 51(D1), D638–D646.

[33] Tijms, B. M., et al. (2024). Cerebrospinal fluid proteomics in patients with Alzheimer's disease reveals five molecular subtypes with distinct genetic risk profiles. *Nature Aging*, 4(1), 33–47.

[34] The UniProt Consortium (2023). UniProt: the universal protein knowledgebase in 2023. *Nucleic Acids Research*, 51(D1), D523–D531.

[35] Välikangas, T., Suomi, T., & Elo, L. L. (2018). A systematic evaluation of normalization methods in quantitative label-free proteomics. *Briefings in Bioinformatics*, 19(1), 1–11.

[36] World Health Organization (2023). *Dementia*. WHO Fact Sheet. Retrieved from https://www.who.int/news-room/fact-sheets/detail/dementia

[37] Zlokovic, B. V. (2011). Neurovascular pathways to neurodegeneration in Alzheimer's disease and other disorders. *Nature Reviews Neuroscience*, 12(12), 723–738.

[38] Zou, H., & Hastie, T. (2005). Regularization and variable selection via the elastic net. *Journal of the Royal Statistical Society: Series B*, 67(2), 301–320.